

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems

COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO3: Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity. (BL2, BL3)

Activity Tool : Experiments (High)

Assessment Tool Weightage: 40%

Dr.G.Ayyappan

QUESTIONS Total Marks: 50			
Q.No	Question	Bloom's Level	Marks
1	For the relation STUDENT_COURSE(SID, SName, CID, CName, Instructor, Grade), identify all functional dependencies and determine the candidate keys.	BL2 – Understand	5
2	Demonstrate the process of converting an unnormalized relation into 1NF with step-by-step justification and example.	BL3 – Apply	5
3	Given relation R(A,B,C,D,E) with FDs: $A \rightarrow BC$, $BC \rightarrow D$, $D \rightarrow E$. Analyze and decompose it to 2NF eliminating partial dependencies.	BL3 – Apply	5
4	Apply 3NF decomposition on the relation EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID) with transitive dependencies.	BL3 – Apply	5
5	Verify whether R(A,B,C,D) with FDs: $AB \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$ is in BCNF. If not, decompose it into BCNF.	BL3 – Apply	5

6	Experimentally verify lossless-join decomposition for a given relation using the tabular (Chase) algorithm.	BL3 – Apply	5
7	Identify the multi-valued dependencies in TEACHER(TID, Subject, Hobby) and decompose it into 4NF.	BL3 – Apply	5

8	Demonstrate how join dependencies lead to redundancy. Decompose a given relation to 5NF (Project-Join Normal Form).	BL3 – Apply	5
9	Given a poorly designed database schema for a college, identify all anomalies (insertion, deletion, update) and normalize it to 3NF.	BL2 – Understand	5
10	Compute the closure of attribute sets for the given FDs and determine all candidate keys for the relation.	BL3 – Apply	5

Code of Conduct:

I **KESHAVARAM L/192511523** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

L. Keshavaram

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts

Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

QUESTION 1 — Functional Dependencies & Candidate Keys

For STUDENT_COURSE(SID, SName, CID, CName, Instructor, Grade), identify all functional dependencies and determine the candidate keys.

2.1 Relation analysis

Assume each student has a unique SID, each course has a unique CID, the instructor assigned to a course is determined by CID, and Grade is determined by a particular student taking a particular course.

STUDENT_COURSE(SID, SName, CID, CName, Instructor, Grade)

2.2 Functional dependencies

- $SID \rightarrow SName$ — a student ID identifies the student's name.
- $CID \rightarrow CName$ — a course ID identifies the course name.
- $CID \rightarrow Instructor$ — a course ID identifies its assigned instructor.
- $(SID, CID) \rightarrow Grade$ — a student's grade depends on the student-course combination.

2.3 Candidate key derivation

Consider the closure of $\{SID, CID\}$:

$$\{SID, CID\}^+ = \{SID, CID, SName, CName, Instructor, Grade\}$$

Therefore, $\{SID, CID\}$ determines every attribute of the relation. Neither SID nor CID alone determines all attributes, so the composite key is minimal.

Candidate key = $\{SID, CID\}$. The relation contains partial dependencies $SID \rightarrow SName$ and $CID \rightarrow CName$, Instructor with respect to the composite key, which is why normalization is required.

2.4 Rubric-oriented analysis

- Concept understanding: distinguishes determinant attributes from dependent attributes.
- Experimental design: uses attribute closure to validate the key.
- Analysis: identifies partial dependencies that create redundancy.
- Documentation: presents FDs, closure and conclusion explicitly.

QUESTION 2 — Unnormalized Relation to 1NF

Demonstrate the process of converting an unnormalized relation into 1NF with step-by-step justification and example.

3.1 Unnormalized form (UNF)

Consider a student record in which a student may register for several courses. A single row contains a repeating group of course details:

SID	SName	Courses
S101	Arun	{(C01, DBMS, A, 85), (C02, OS, B, 78)}
S102	Meena	{(C01, DBMS, A, 91)}

The Courses field contains multiple values in one attribute. Therefore, the relation is not in 1NF because attribute values are not atomic.

3.2 Conversion to 1NF

1. Identify the repeating group: course ID, course name, instructor and grade.
2. Remove the repeating group from the single cell structure.
3. Create one row for each student-course occurrence.
4. Ensure every attribute contains one atomic value.
5. Use the composite key (SID, CID) to uniquely identify each enrollment.

SID	SName	CID	CName	Instructor	Grade
S101	Arun	C01	DBMS	A	85
S101	Arun	C02	OS	B	78
S102	Meena	C01	DBMS	A	91

Every cell now contains exactly one atomic value, there are no repeating groups, and each row represents one student-course enrollment.

The UNF relation has been converted into 1NF by eliminating repeating groups and ensuring atomic attribute values. Redundancy may still exist; 1NF alone does not guarantee 2NF or 3NF.

QUESTION 3 — 2NF Analysis and Decomposition

Given $R(A,B,C,D,E)$ with FDs $A \rightarrow BC$, $BC \rightarrow D$, $D \rightarrow E$, analyze and decompose it to 2NF eliminating partial dependencies.

4.1 Given functional dependencies

- $A \rightarrow B, C$
- $BC \rightarrow D$
- $D \rightarrow E$

4.2 Candidate key

$$A^+ = \{A\} \rightarrow \{A,B,C\} \rightarrow \{A,B,C,D\} \rightarrow \{A,B,C,D,E\}$$

Thus A is a candidate key because A determines every attribute and is already a single attribute.

4.3 2NF test

Second Normal Form requires the relation to be in 1NF and to have no partial dependency of a non-prime attribute on a proper subset of a candidate key. Here the candidate key contains only one attribute, A. A single-attribute key has no proper non-empty subset.

There are NO partial dependencies to eliminate. Therefore, the given relation is already in 2NF (assuming 1NF). No decomposition is required for the specific 2NF condition.

2NF relation	Status
$R(A,B,C,D,E)$	Already in 2NF

However, the dependency $D \rightarrow E$ is a transitive dependency through $A \rightarrow \dots \rightarrow D$, so the relation is not necessarily in 3NF. This is a separate issue and should not be incorrectly treated as a 2NF violation.

$R(A,B,C,D,E)$ is already in 2NF because its candidate key A is not composite. The correct 2NF decomposition is therefore the unchanged relation R itself.

QUESTION 4 — 3NF Decomposition

Apply 3NF decomposition on $EMPLOYEE(EMPID, EMPNAME, DEPTID, DEPTNAME, MANAGERID)$ with transitive dependencies.

5.1 Functional dependencies

- $\text{EmpID} \rightarrow \text{EmpName}, \text{DeptID}$
- $\text{DeptID} \rightarrow \text{DeptName}, \text{ManagerID}$

$$\text{EmpID} \rightarrow \text{DeptID} \rightarrow \{\text{DeptName}, \text{ManagerID}\}$$

The dependency of DeptName and ManagerID on EmpID occurs transitively through DeptID. This creates update and redundancy problems.

5.2 3NF decomposition

Relation	Attributes	Primary Key
EMPLOYEE	EmpID, EmpName, DeptID	EmpID
DEPARTMENT	DeptID, DeptName, ManagerID	DeptID

The decomposition separates employee-specific facts from department-specific facts. DeptID in EMPLOYEE acts as a foreign key referencing DEPARTMENT(DeptID).

```
CREATE TABLE DEPARTMENT (  
  DeptID INT PRIMARY KEY,  
  DeptName VARCHAR(100),  
  ManagerID INT  
);  
  
CREATE TABLE EMPLOYEE (  
  EmpID INT PRIMARY KEY,  
  EmpName VARCHAR(100),  
  DeptID INT,  
  FOREIGN KEY (DeptID) REFERENCES DEPARTMENT(DeptID)  
);
```

The decomposition removes the transitive dependency from the employee relation and produces a 3NF design under the stated dependencies.

QUESTION 5 — BCNF Verification & Decomposition

Verify whether $R(A,B,C,D)$ with FDs $AB \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$ is in BCNF. If not, decompose it into BCNF.

6.1 Candidate keys

- $AB^+ = \{A, B, C, D\}$ because $AB \rightarrow C$ and $C \rightarrow D$.
- $BC^+ = \{B, C, D, A\}$ because $C \rightarrow D$ and $D \rightarrow A$.
- $BD^+ = \{B, D, A, C\}$ because $D \rightarrow A$ and $AB \rightarrow C$.

Candidate keys are therefore AB, BC and BD.

6.2 BCNF test

For every non-trivial FD $X \rightarrow Y$, BCNF requires X to be a superkey.

FD	Is determinant a superkey?	BCNF status
$AB \rightarrow C$	Yes; AB is a candidate key	Satisfies
$C \rightarrow D$	No; C is not a superkey	Violates BCNF
$D \rightarrow A$	No; D is not a superkey	Violates BCNF

6.3 Decomposition

Use the violating dependency $C \rightarrow D$.

Resulting relation	Attributes	Key
R1	C, D	C
R2	A, B, C	AB

R1 is in BCNF because $C \rightarrow D$ and C is a key of R1. In R2, $AB \rightarrow C$ and AB is a key; the problematic dependencies $C \rightarrow D$ and $D \rightarrow A$ are no longer dependencies within R2.

The original relation is not in BCNF. A valid BCNF decomposition is R1(C,D) and R2(A,B,C).

QUESTION 6 — Lossless-Join Decomposition Using Chase

Experimentally verify lossless-join decomposition for a given relation using the tabular (Chase) algorithm.

7.1 Worked experimental setup

Because the assessment brief does not specify a particular relation/decomposition for this question, the following explicit example is used to demonstrate the Chase procedure.

Let $R(A, B, C)$ with functional dependency $B \rightarrow C$ and decomposition:

R1(A,B) and R2(B,C)

7.2 Initial Chase table

Row	A	B	C
R1	a1	b1	c*
R2	a*	b1	c1

The common attribute between R1 and R2 is B. Both rows contain the same distinguished symbol b1 for B. The FD $B \rightarrow C$ therefore forces the C value in the first row to agree with the distinguished C value in the second row.

7.3 Chase step

Row	A	B	C
R1 after applying $B \rightarrow C$	a1	b1	c1
R2	a*	b1	c1

A row becomes fully distinguished in the attributes of R because it contains the distinguished symbols for A, B and C. Hence the decomposition is lossless with respect to the stated dependency.

7.4 Interpretation

- The Chase starts from a tableau representing the decomposed relations.
- Functional dependencies are repeatedly applied to enforce equality constraints.
- If a row becomes a fully distinguished row, the decomposition is lossless.
- The experiment demonstrates how dependencies can guarantee that joining the decomposed relations does not create spurious tuples.

The decomposition $R(A,B,C) \rightarrow R1(A,B), R2(B,C)$ is lossless for the FD $B \rightarrow C$, as verified by the Chase tableau.

QUESTION 7 — 4NF Using Multi-Valued Dependencies

Identify the multi-valued dependencies in TEACHER(TID, Subject, Hobby) and decompose it into 4NF.

8.1 MVD identification

Assume that a teacher can teach several independent subjects and can have several independent hobbies. Subjects and hobbies are independent multi-valued facts about a teacher.

$$TID \twoheadrightarrow Subject$$

$$TID \twoheadrightarrow Hobby$$

The relation stores combinations of every subject with every hobby for the same teacher, which can produce redundant tuples.

8.2 Example redundancy

TID	Subject	Hobby
T01	DBMS	Cricket
T01	DBMS	Music
T01	OS	Cricket
T01	OS	Music

The four rows represent two independent subject values and two independent hobby values. Storing their Cartesian combinations creates multivalued redundancy.

8.3 4NF decomposition

Relation	Attributes	Key
TEACHER_SUBJECT	TID, Subject	(TID, Subject)
TEACHER_HOBBY	TID, Hobby	(TID, Hobby)

The original relation is decomposed into TEACHER_SUBJECT(TID, Subject) and TEACHER_HOBBY(TID, Hobby), removing the independent multi-valued facts and satisfying the 4NF design goal.

QUESTION 8 — Join Dependency & 5NF

Demonstrate how join dependencies lead to redundancy. Decompose a given relation to 5NF (Project-Join Normal Form).

9.1 Worked example

Because the assessment brief does not specify the attributes for the relation in this question, use the standard Supplier–Part–Project example:

SPJ(Supplier, Part, Project)

Suppose a supplier can supply a part, a supplier can work on a project, and a part can be used in a project independently. The relation may need to store combinations implied by three pairwise relationships.

9.2 Redundancy due to join dependency

Supplier	Part	Project
S1	P1	J1
S1	P1	J2
S1	P2	J1
S1	P2	J2

If these combinations are implied by the pairwise facts SP, SJ and PJ, storing every triple duplicates information. This corresponds to a non-trivial join dependency.

$*(SP, SJ, PJ)$

9.3 5NF decomposition

Relation	Attributes	Meaning
SP	Supplier, Part	Supplier supplies part
SJ	Supplier, Project	Supplier works on project
PJ	Part, Project	Part is used for project

The original SPJ relation can be reconstructed by joining SP, SJ and PJ when the join dependency is valid. The decomposition removes redundancy caused by the non-trivial join dependency.

The relation is decomposed into SP(Supplier,Part), SJ(Supplier,Project), and PJ(Part,Project), illustrating the core idea of 5NF: eliminate non-trivial join dependencies that cause redundancy.

QUESTION 9 — Anomalies & 3NF Normalization

Given a poorly designed database schema for a college, identify all anomalies (insertion, deletion, update) and normalize it to 3NF.

10.1 Poorly designed schema

For a complete worked example, consider:

COLLEGE_RECORD(SID, SName, DeptID, DeptName, CID, CName, Instructor,

Grade) Assume the following dependencies:

- $SID \rightarrow SName, DeptID$
- $DeptID \rightarrow DeptName$
- $CID \rightarrow CName, Instructor$
- $(SID, CID) \rightarrow Grade$

10.2 Anomalies

Anomaly	Explanation
Update anomaly	Changing a department name requires updating many student-course rows; missing one row creates inconsistent data.
Insertion anomaly	A new department or course may be difficult to insert if no student enrollment exists yet.
Deletion anomaly	Deleting the last enrollment for a course or department can accidentally remove the only stored information about that course/department.

10.3 3NF decomposition

Relation	Attributes	Primary Key
STUDENT	SID, SName, DeptID	SID
DEPARTMENT	DeptID, DeptName	DeptID
COURSE	CID, CName, Instructor	CID
ENROLLMENT	SID, CID, Grade	(SID, CID)

10.4 Dependency placement

- STUDENT stores student-specific information and references DEPARTMENT.
- DEPARTMENT stores department-specific information.
- COURSE stores course-specific information.
- ENROLLMENT stores the relationship between a student and a course plus the Grade.

```

CREATE TABLE DEPARTMENT (
  DeptID INT PRIMARY KEY,
  DeptName VARCHAR(100)
);

CREATE TABLE STUDENT (
  SID INT PRIMARY KEY,
  SName VARCHAR(100),
  DeptID INT,
  FOREIGN KEY (DeptID) REFERENCES DEPARTMENT(DeptID)
);

CREATE TABLE COURSE (
  CID INT PRIMARY KEY,
  CName VARCHAR(100),
  Instructor VARCHAR(100)
);

CREATE TABLE ENROLLMENT (
  SID INT,
  CID INT,
  Grade VARCHAR(5),
  PRIMARY KEY (SID, CID),
  FOREIGN KEY (SID) REFERENCES STUDENT(SID),
  FOREIGN KEY (CID) REFERENCES COURSE(CID)
);

```

The decomposition separates independent facts, removes transitive dependencies and reduces insertion, deletion and update anomalies. The resulting design is in 3NF under the stated dependencies.

QUESTION 10 — Attribute Closure & Candidate Keys

Compute the closure of attribute sets for the given FDs and determine all candidate keys for the relation.

The assessment brief does not provide the specific relation and FD set for Q10. To demonstrate the required calculation completely, this submission uses $R(A,B,C,D,E)$ with FDs: $A \rightarrow B$, $B \rightarrow C$, $CD \rightarrow E$, $E \rightarrow A$.

11.1 Closure calculations

Given $F = \{A \rightarrow B, B \rightarrow C, CD \rightarrow E, E \rightarrow A\}$:

$$A^+ = \{A, B, C\}$$

A alone cannot derive D, so A is not a candidate key.

$$AD^+ = \{A, D\} \rightarrow \{A, B, D\} \rightarrow \{A, B, C, D\} \rightarrow \{A, B, C, D, E\}$$

Therefore AD is a candidate key.

$$BD^+ = \{B, D\} \rightarrow \{B, C, D\} \rightarrow \{B, C, D, E\} \rightarrow \{A, B, C, D, E\}$$

Therefore BD is a candidate key.

$$CD^+ = \{C, D\} \rightarrow \{C, D, E\} \rightarrow \{A, C, D, E\} \rightarrow \{A, B, C, D, E\}$$

Therefore CD is a candidate key.

$$DE^+ = \{D, E\} \rightarrow \{A, D, E\} \rightarrow \{A, B, D, E\} \rightarrow \{A, B, C, D, E\}$$

Therefore DE is a candidate key.

11.2 Candidate key verification

Set	Closure	Candidate key?
AD	$\{A, B, C, D, E\}$	Yes
BD	$\{A, B, C, D, E\}$	Yes
CD	$\{A, B, C, D, E\}$	Yes
DE	$\{A, B, C, D, E\}$	Yes

Each listed key is minimal: removing either attribute from AD, BD, CD or DE prevents the closure from reaching all attributes.

11.3 General closure algorithm

- Start with $X^+ = X$.
- Scan the functional dependencies.
- If the left side of an FD is contained in X^+ , add its right-side attributes.
- Repeat until no new attributes can be added.
- If X^+ contains every attribute of R, X is a superkey.
- Check minimality by removing one attribute at a time; if no proper subset is a superkey, X is a candidate key.

For the worked Q10 example, the candidate keys are AD, BD, CD and DE. The closure method provides a systematic way to validate candidate keys.